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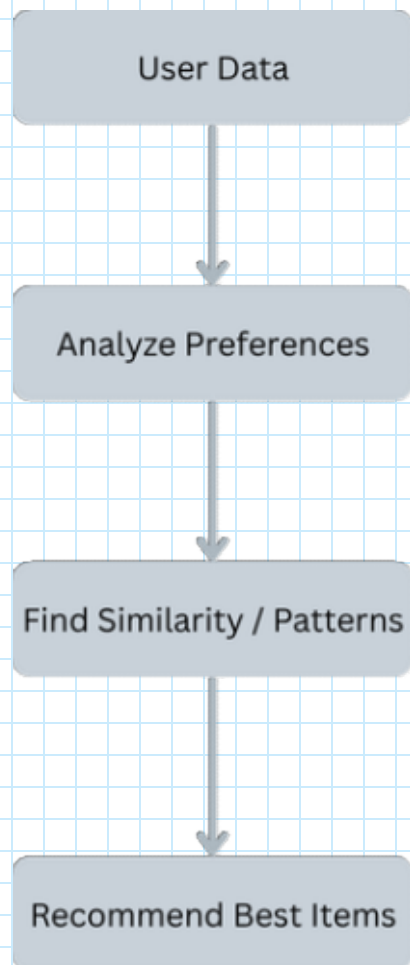
Recommendation Systems

1. What is a Recommendation System?

A **Recommendation System (Recommender System)** is a Machine Learning technique that suggests **products, movies, songs, books, videos, news articles, or other items** to users based on their preferences, behaviors, or similarities.

Its goal is to recommend the **most relevant items** to each user.

Main Idea



2. Why are Recommendation Systems Important?

Modern applications contain millions of products or content items.
Without recommendations:

✗ Users struggle to find relevant items.

With recommendations:

- ☒ Personalized experience
- ☒ Increased customer satisfaction
- ☒ Higher sales
- ☒ Better engagement

3. Real-Life Examples

Netflix

Recommends:

- Movies
- TV Shows

based on watching history.

Amazon

Recommends:

- Products
- Accessories

Example:

Customers who bought this item
also bought...

YouTube

Recommends:

- Videos
- Shorts

based on:

- Watch history
- Search history
- Likes
- Subscriptions

Spotify

Recommends:

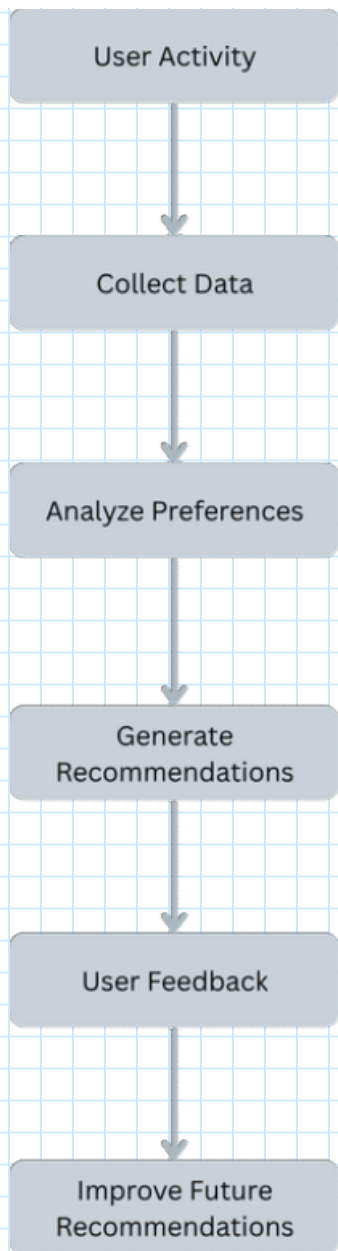
- Songs
- Artists
- Albums

Instagram

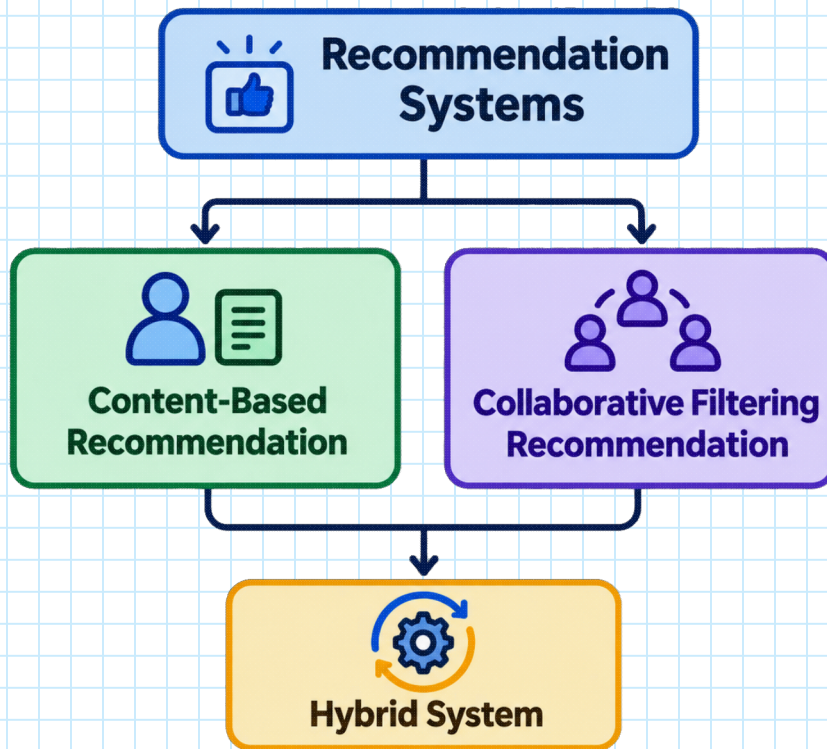
Recommends:

- Reels
- Posts
- Accounts

4. How Recommendation Systems Work



5. Types of Recommendation Systems



The three major approaches are:

- Content-Based Filtering
- Collaborative Filtering
- Hybrid Recommendation System

